



Architecture Big data & applications

Compte rendu

Lab 5: Cluster spark avec docker

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1. Installation Cluster Spark :

```
root@hadoop-master:~# ./start-spark.sh

starting org.apache.spark.deploy.master.Master, logging to /usr/local/spark/logs/spark--org.apache.spark.deploy.master.Master-1-hadoop-master.out
hadoop-slave2: Warning: Permanently added 'hadoop-slave2,172.18.0.2' (ECDSA) to the list of known hosts.
hadoop-slave1: Warning: Permanently added 'hadoop-slave1,172.18.0.3' (ECDSA) to the list of known hosts.
hadoop-slave1: starting org.apache.spark.deploy.worker.Worker, logging to /usr/local/spark/logs/spark-root-org.apache.spark.deploy.worker.Worker-1-hadoop-slave1.out
hadoop-slave2: starting org.apache.spark.deploy.worker.Worker, logging to /usr/local/spark/logs/spark-root-org.apache.spark.deploy.worker.Worker-1-hadoop-slave2.out

root@hadoop-master:~# jps
544 SecondaryNameNode
1251 Master
854 ResourceManager
1355 Jps
253 NameNode
```

2. Premiers exemples sur apache spark :

- Premier exemple spark avec spark-submit

```
root@hadoop-master:~# spark-submit --class org.apache.spark.examples.SparkPi --master local[*] $SPARK_HOME/examples/jars/spark-examples_2.12-3.2.4.jar 100
2025-10-28 21:39:34,859 INFO spark.SparkContext: Running Spark version 3.2.4
2025-10-28 21:39:35,054 INFO resource.ResourceUtils: =====
2025-10-28 21:39:35,054 INFO resource.ResourceUtils: No custom resources configured for spark.driver.
2025-10-28 21:39:35,054 INFO resource.ResourceUtils: =====
2025-10-28 21:39:35,056 INFO spark.SparkContext: Submitted application: Spark Pi
2025-10-28 21:39:35,083 INFO resource.ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, amount: 1, script: , vendor: , memory -> name: memory, amount: 2048, script: , vendor: , offHeap -> name: offHeap, amount: 0, script: , vendor: ), task resources: Map(cpus -> name: cpus, amount: 1.0)
2025-10-28 21:39:35,100 INFO resource.ResourceProfile: Limiting resource is cpus at 1 tasks per executor
2025-10-28 21:39:35,103 INFO resource.ResourceProfileManager: Added ResourceProfile id: 0
2025-10-28 21:39:35,167 INFO spark.SecurityManager: Changing view acls to: root
2025-10-28 21:39:35,167 INFO spark.SecurityManager: Changing modify acls to: root
2025-10-28 21:39:35,168 INFO spark.SecurityManager: Changing view acls groups to:
2025-10-28 21:39:35,168 INFO spark.SecurityManager: Changing modify acls groups to:
2025-10-28 21:39:35,169 INFO spark.SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view permissions: Set(root); groups with view permissions: Set(); users with modify permissions: Set(root); groups with modify permissions: Set()
2025-10-28 21:39:35,427 INFO util.Utils: Successfully started service 'sparkDriver' on port 43669.
2025-10-28 21:39:35,463 INFO spark.SparkEnv: Registering MapOutputTracker
2025-10-28 21:39:35,505 INFO spark.SparkEnv: Registering BlockManagerMaster
2025-10-28 21:39:35,532 INFO storage.BlockManagerMasterEndpoint: Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
2025-10-28 21:39:35,532 INFO storage.BlockManagerMasterEndpoint: BlockManagerMasterEndpoint up
2025-10-28 21:39:35,539 INFO spark.SparkEnv: Registering BlockManagerMasterHeartbeat
2025-10-28 21:39:35,568 INFO storage.DiskBlockManager: Created local directory at /tmp/blockmgr-0eb9f6a0-d8eb-4e9f-9575-edc320c40024
2025-10-28 21:39:35,593 INFO memory.MemoryStore: MemoryStore started with capacity 2004.6 MiB
2025-10-28 21:39:35,612 INFO spark.SparkEnv: Registering OutputCommitCoordinator

2025-10-28 21:39:46,931 INFO executor.Executor: Finished task 90.0 in stage 0.0 (TID 90). 957 bytes result sent to driver
2025-10-28 21:39:46,932 INFO scheduler.TaskSetManager: Finished task 90.0 in stage 0.0 (TID 90) in 1065 ms on hadoop-master (executor driver) (97/100)
2025-10-28 21:39:46,935 INFO executor.Executor: Finished task 98.0 in stage 0.0 (TID 98). 957 bytes result sent to driver
2025-10-28 21:39:46,935 INFO executor.Executor: Finished task 97.0 in stage 0.0 (TID 97). 957 bytes result sent to driver
2025-10-28 21:39:46,936 INFO scheduler.TaskSetManager: Finished task 98.0 in stage 0.0 (TID 98) in 502 ms on hadoop-master (executor driver) (98/100)
2025-10-28 21:39:46,936 INFO scheduler.TaskSetManager: Finished task 97.0 in stage 0.0 (TID 97) in 649 ms on hadoop-master (executor driver) (99/100)
2025-10-28 21:39:46,937 INFO executor.Executor: Finished task 99.0 in stage 0.0 (TID 99). 957 bytes result sent to driver
2025-10-28 21:39:46,938 INFO scheduler.TaskSetManager: Finished task 99.0 in stage 0.0 (TID 99) in 489 ms on hadoop-master (executor driver) (100/100)
2025-10-28 21:39:46,938 INFO scheduler.TaskSchedulerImpl: Removed TaskSet 0.0, whose tasks have all completed, from pool
2025-10-28 21:39:46,940 INFO scheduler.DAGScheduler: ResultStage 0 (reduce at SparkPi.scala:38) finished in 8.448 s
2025-10-28 21:39:46,947 INFO scheduler.DAGScheduler: Job 0 is finished. Cancelling potential speculative or zombie tasks for this job
2025-10-28 21:39:46,947 INFO scheduler.TaskSchedulerImpl: Killing all running tasks in stage 0: Stage finished
2025-10-28 21:39:46,950 INFO scheduler.DAGScheduler: Job 0 finished: reduce at SparkPi.scala:38, took 8.607421 s
Pi is roughly 3.1427439142743916
2025-10-28 21:39:46,966 INFO server.AbstractConnector: Stopped Spark@7561db12{HTTP/1.1, (http/1.1)}{0.0.0.0:4040}
2025-10-28 21:39:46,970 INFO ui.SparkUI: Stopped Spark web UI at http://hadoop-master:4040
2025-10-28 21:39:46,985 INFO spark.MapOutputTrackerMasterEndpoint: Deleting directory /tmp/spark-2713240a-c2c6-444b-b3ce-70ecc5924125
2025-10-28 21:39:47,001 INFO memory.MemoryStore: MemoryStore cleared
2025-10-28 21:39:47,002 INFO storage.BlockManager: BlockManager stopped
2025-10-28 21:39:47,010 INFO storage.BlockManagerMaster: BlockManagerMaster stopped
2025-10-28 21:39:47,013 INFO scheduler.OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
2025-10-28 21:39:47,023 INFO spark.SparkContext: Successfully stopped SparkContext
2025-10-28 21:39:47,027 INFO util.ShutdownHookManager: Shutdown hook called
2025-10-28 21:39:47,028 INFO util.ShutdownHookManager: Deleting directory /tmp/spark-2713240a-c2c6-444b-b3ce-70ecc5924125
2025-10-28 21:39:47,033 INFO util.ShutdownHookManager: Deleting directory /tmp/spark-8ac6d5c1-d6c4-4a86-b2b4-0ccb31581387
root@hadoop-master:~#
```

- Application word count avec spark-shell

[illegible]

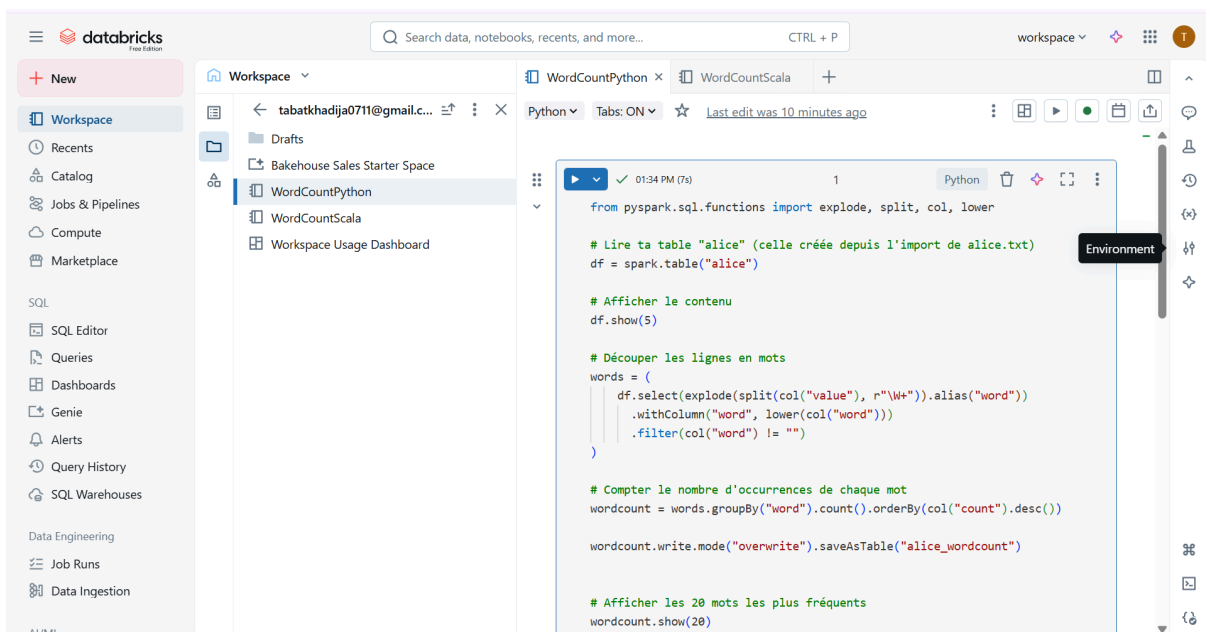
- Soumettre une Application python :

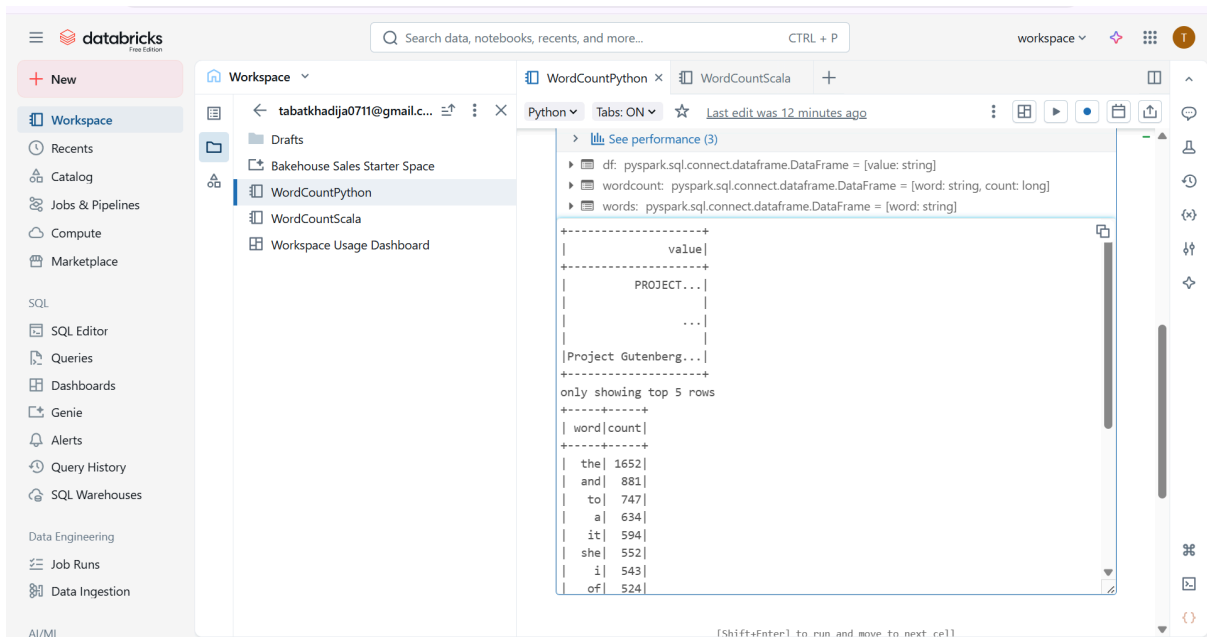
```

root@hadoop-master: ~
root@hadoop-master:~# cat > /root/wordcount.py <<'PY'
> # /root/wordcount.py
> from pyspark.sql import SparkSession
>
> spark = SparkSession.builder \
>     .master("yarn") \
>     .appName("wordcount") \
>     .getOrCreate()
>
> sc = spark.sparkContext
>
> data = sc.textFile("hdfs://hadoop-master:9000/user/root/input/alice.txt")
> words = data.flatMap(lambda line: line.split(" "))
> wordCounts = words.map(lambda w: (w, 1)).reduceByKey(lambda a, b: a + b)
> wordCounts.saveAsTextFile("hdfs://hadoop-master:9000/user/root/output/rr2")
>
> spark.stop()
>
> PY
root@hadoop-master:~# ls
alice.txt                start-jupyter.sh
fairscheduler-statedump.log start-kafka-zookeeper.sh
hdfs                    start-spark.sh
run-wordcount.sh        wordcount.py
start-hadoop.sh
root@hadoop-master:~# ls -l /root/wordcount.py
-rw-r--r-- 1 root root 476 Oct 29 14:08 /root/wordcount.py
root@hadoop-master:~# cat -n /root/wordcount.py
 1 # /root/wordcount.py
 2 from pyspark.sql import SparkSession
 3
 4 spark = SparkSession.builder \
 5     .master("yarn") \
 6     .appName("wordcount") \
 7     .getOrCreate()
 8

```

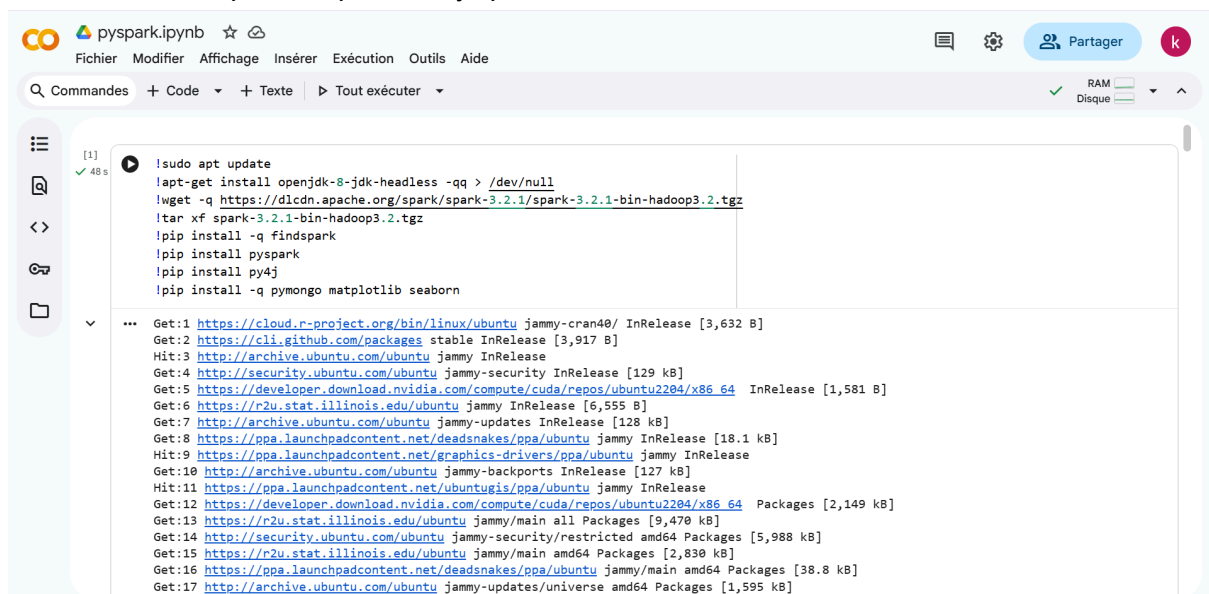
- Un exemple d' une Application WordCount en python sur databricks :





3. Installer pyspark sur colab :

- Installer Apache Spark et PySpark



- Configuration de l'environnement et démarrage d'une session Spark

```
[4]
✓ 0 s
import os
import sys
import findspark
# os.environ["JAVA_HOME"] = "/usr/lib/jvm/java-11-openjdk-amd64"
# os.environ["SPARK_HOME"] = "/content/spark-3.5.1-bin-hadoop3.2"
findspark.init()
findspark.find()

' /usr/local/lib/python3.12/dist-packages/pyspark'

[6]
✓ 0 s
import pyspark
from pyspark.sql import SparkSession
spark = SparkSession.builder \
    .appName("ColabSpark") \
    .config("spark.driver.memory", "2g") \
    .getOrCreate()
print(" Spark est configuré avec succès !")

Spark est configuré avec succès !
```

- Premier exemple

pyspark.ipynb ☆ ☁

Fichier Modifier Affichage Insérer Exécution Outils Aide

Commandes + Code + Texte ▶ Tout exécuter

```
[7]
✓ 0 s
data = [(1, "Alice", 23), (2, "Bob", 30), (3, "Charlie", 29)]
columns = ["id", "nom", "age"]
df = spark.createDataFrame(data, columns)

[8]
✓ 2 s
df.show()

+---+-----+---+
| id|    nom|age|
+---+-----+---+
|  1|  Alice| 23|
|  2|    Bob| 30|
|  3|Charlie| 29|
+---+-----+---+

[9]
✓ 1 s
df.printSchema() # Structure du DataFrame
df.select("nom", "age").show() # Sélection de colonnes
df.filter(df.age > 25).show() # Filtrage des données

... root
|-- id: long (nullable = true)
|-- nom: string (nullable = true)
|-- age: long (nullable = true)

+-----+-----+
```

- Chargement et manipulation des données avec Spark

pyspark.ipynb

Fichier Modifier Affichage Insérer Exécution Outils Aide

Commandes + Code + Texte Tout exécuter

RAM Disque

```
[10]
✓ 3 s
df = spark.read.csv("/transaction_data.csv", header=True, inferSchema=True)
df.show(5)
```

Transaction ID	Sender Account ID	Receiver Account ID	Transaction Amount	Transaction Type	Timestamp	Transaction Status	Fraud Flag
TXN9520068950	ACC14994	ACC16656	495.9	Deposit	2025-01-17 10:14:00	Failed	true
TXN9412011085	ACC58958	ACC32826	529.62	Withdrawal	2025-01-17 10:51:00	Success	false
TXN4407425052	ACC56321	ACC92481	862.47	Withdrawal	2025-01-17 10:50:00	Failed	false
TXN2214150284	ACC48650	ACC76457	1129.88	Transfer	2025-01-17 10:56:00	Success	true
TXN4247571145	ACC60921	ACC11419	933.24	Deposit	2025-01-17 10:25:00	Success	true

only showing top 5 rows

```
[11]
✓ 0 s
df.printSchema()
```

```
root
|-- Transaction ID: string (nullable = true)
|-- Sender Account ID: string (nullable = true)
|-- Receiver Account ID: string (nullable = true)
|-- Transaction Amount: double (nullable = true)
|-- Transaction Type: string (nullable = true)
|-- Timestamp: timestamp (nullable = true)
|-- Transaction Status: string (nullable = true)
|-- Fraud Flag: boolean (nullable = true)
```

pyspark.ipynb

Fichier Modifier Affichage Insérer Exécution Outils Aide

Commandes + Code + Texte Tout exécuter

RAM Disque

```
-- Device Used: string (nullable = true)
-- Network Slice ID: string (nullable = true)
-- Latency (ms): integer (nullable = true)
-- Slice Bandwidth (Mbps): integer (nullable = true)
-- PIN Code: integer (nullable = true)
```

```
[12]
✓ 0 s
df.filter(df["Transaction Amount"] > 1000).show()
```

Transaction ID	Sender Account ID	Receiver Account ID	Transaction Amount	Transaction Type	Timestamp	Transaction Status	Fraud Flag
TXN2214150284	ACC48650	ACC76457	1129.88	Transfer	2025-01-17 10:56:00	Success	true
TXN3109277527	ACC48227	ACC93536	1135.8	Deposit	2025-01-17 10:56:00	Failed	true
TXN4541918858	ACC10284	ACC24018	1154.59	Withdrawal	2025-01-17 10:38:00	Success	false
TXN2166128709	ACC11156	ACC57086	1356.77	Deposit	2025-01-17 10:06:00	Failed	false
TXN6682052855	ACC68764	ACC53807	1171.23	Transfer	2025-01-17 10:29:00	Success	false
TXN5541586171	ACC72546	ACC73085	1196.23	Withdrawal	2025-01-17 10:48:00	Failed	false
TXN2415439100	ACC92329	ACC71125	1414.14	Transfer	2025-01-17 10:30:00	Failed	false
TXN3884612815	ACC79800	ACC11375	1453.39	Withdrawal	2025-01-17 10:12:00	Failed	false
TXN7417651729	ACC26106	ACC21043	1371.8	Transfer	2025-01-17 10:09:00	Failed	true
TXN6661143500	ACC94124	ACC34396	1227.89	Transfer	2025-01-17 10:29:00	Success	false
TXN2465139688	ACC21498	ACC64106	1483.75	Transfer	2025-01-17 10:03:00	Success	false
TXN6050985831	ACC90300	ACC78072	1093.14	Withdrawal	2025-01-17 10:05:00	Failed	false
TXN2157994961	ACC60872	ACC56165	1017.3	Deposit	2025-01-17 10:52:00	Failed	true
TXN2467998038	ACC92520	ACC54323	1436.77	Withdrawal	2025-01-17 11:00:00	Failed	false
TXN4014434434	ACC50922	ACC98669	1094.34	Transfer	2025-01-17 10:55:00	Failed	false
TXN5548725719	ACC49858	ACC26440	1235.27	Deposit	2025-01-17 10:31:00	Success	false

```
[13]
✓ 1 s
df.groupBy("Transaction Type").sum("Transaction Amount").show()
```

Transaction Type	sum(Transaction Amount)
Deposit	252042.61999999998
Transfer	291776.55000000002
Withdrawal	227346.12000000002

```
[14]
df.orderBy(df["Transaction Amount"].desc()).show(5)
```

Transaction ID	Sender Account ID	Receiver Account ID	Transaction Amount	Transaction Type	Timestamp	Transaction Status	Fraud Flag
TXN1696527060	ACC44804	ACC26269	1497.76	Transfer	2025-01-17 10:54:00	Success	true
TXN8741905833	ACC67128	ACC77786	1495.01	Transfer	2025-01-17 10:46:00	Success	true
TXN4979176158	ACC76875	ACC22724	1494.2	Deposit	2025-01-17 10:31:00	Failed	true
TXN7486103501	ACC14340	ACC33467	1494.03	Transfer	2025-01-17 10:07:00	Success	true
TXN4433757533	ACC47545	ACC63108	1494.01	Transfer	2025-01-17 10:28:00	Success	false

only showing top 5 rows